THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the radius of the Earth is about 6400 km, the thickness of the crust (about 35 km) is: 1

(a) less than 1 per cent of the radius **(b)** about 25 per cent of the radius
(c) about 50 per cent of the radius **(d)** equal to the radius

2. If air contains 21 per cent oxygen, in 500 L of air the volume of oxygen is about: 1

(a) 210 L **(b)** 78 L **(c)** 105 L **(d)** 21 L

3. The percentage of Earth's water that is fresh water is about: 1

(a) 97 per cent **(b)** 30 per cent **(c)** 3 per cent **(d)** 50 per cent

4. Venus has a surface temperature of about 460 degrees Celsius mainly due to: 1

(a) closeness to the Moon **(b)** its oceans
(c) a thick CO_2 atmosphere **(d)** its magnetic field

5. The inner core is solid even though it is very hot because of: 1

(a) extremely high pressure **(b)** no gravity
(c) low temperature **(d)** water

6. In 200 L of air the volume of nitrogen is about: 1

(a) 78 L **(b)** 42 L **(c)** 156 L **(d)** 21 L

7. The Earth's water is about 97 per cent salt water. If total water is 1000 units, the units of fresh water are about: 1

(a) 300 **(b)** 970 **(c)** 30 **(d)** 3

8. Mars is cold because it: 1

(a) is nearer to the Sun
(b) has oceans
(c) has a thick atmosphere
(d) is farther from the Sun and has a thin atmosphere

9. Scientists get information about the interior of the Earth from: 1

(a) photographs of the core **(b)** digging to the core
(c) seismic waves of earthquakes **(d)** telescopes

10. At higher altitudes, breathing is difficult because: 1

(a) oxygen is poisonous
(b) air is thinner with lower oxygen amount per breath
(c) air is thicker
(d) air is warm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the inner core solid though the temperature is very high? 2

12. Why does the temperature fall with height in the troposphere but rise in the stratosphere? 2

13. Why do mountaineers carry oxygen cylinders? 2

14. Calculate the volume of oxygen in 250 L of air (21 per cent). 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The Earth's radius is 6400 km. The crust is 35 km, mantle 2900 km. Find the total thickness of the core. 3

16. Explain why water exists in all three states on Earth but not on the Moon. 3

17. Explain why the sky appears blue during the day from the surface of the Earth. 3

OR

Explain why the layers of the Earth are arranged by density with the heaviest at the centre. 3

18. State the effects of melting of glaciers on rivers, sea level and coastal regions. 3

OR

Describe how the Earth's atmosphere and oceans exchange gases, with CO_2 as an example. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the structure of the Earth's interior with the thickness (approximately), state and composition of each layer, and how it is studied. 5

OR

Explain the causes and effects of ozone depletion, mention the gases responsible, and the steps taken by the world to repair the layer. 5

20. Explain the structure of the atmosphere with heights, features and importance of each layer. 5

OR

Compare the conditions on the Earth, Moon, Mars and Venus in a table of five features. 5

END OF QUESTION PAPER Total: 40 Marks

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